IIIT, Bhubaneswar

ET124: Fundamentals of Image Processing

(7^{th} Sem. ETC)

Full Marks: 30

Mid Semester Examination

Duration: 90 mins.

(Answer any five questions including Q1. All questions carry equal marks)

1. (a) Prove that 1D DFT transform matrix is unitary and symmetric. [2×3=6]
   (b) What does the standard deviation of a histogram tell us about the image?
   (c) Determine a 5×5 Gaussian kernel using PASCAL's triangle method.

2. Explain the type of images formed based on the radiation from EM spectrum. [6]

3. (a) A 4×4, 4-bits/pixel image f(m,n) is passed through point wise intensity transformation $g(m,n) = \text{round}\,(10 \times \sqrt{f(m,n)}\,)$. Determine the output image g(m,n) when [4+2=6]

$$f(m,n) = \begin{bmatrix} 12 & 8 & 4 & 9 \\ 10 & 5 & 3 & 6 \\ 8 & 12 & 9 & 13 \\ 4 & 12 & 9 & 10 \end{bmatrix}.$$

   (b) Find the four bit planes of the given image f(m,n).

4. (a) Explain m-adjacency and its advantage with an example. [3+3=6]

   (b) Consider the image segment

$$\begin{array}{cccccc} & 3 & 1 & 2 & 1 & (q) \\ & 2 & 2 & 0 & 2 & \\ & 1 & 2 & 1 & 1 & \\ (p) & 1 & 0 & 1 & 2 & \end{array}$$

   . Let V = {1,2} and compute the length of the shortest 4-, 8-, and m-path between p and q.

5. Explain the first-order derivative and second-order derivative image sharpening filters. [3+3=6]

6. Explain the Hadamard Transform. Generate the transformation matrix for N=8 using mathematical equation and also using kronekar product method. [4+2=6]